RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Date:	**Test - 3**	**Max. Marks:60**
Semester:6th	**UG**	**Duration: 2 Hrs**
Course Title: Cryptography & Network Security		**Course Code:CS365TDF**

Department of CSE

S No	Questions Part- A	M	BT	CO
1.	The public key in DSA is derived from the private key and a _____ generator value.	1	1	1
2.	One method of public key distribution is using a trusted third party called a ----------	1	1	3
3.	The standard format for public key certificates is defined by ----------	1	2	2
4.	The central authority in a Kerberos system is the ----------	1	1	1
5.	Kerberos issues a special ticket known as a ----------	1	1	2
6.	In SSL/TLS, data is divided into blocks and encrypted using _____ key cryptography.	1	1	2
7.	SSL operates between the application layer and the _____ layer in the OSI model.	1	1	3
8.	The two main IPSec protocols are ---------- and ----------	2	1	3
9.	The _____ protocol in 802.11i provides authentication and encryption for wireless communication.	1	1	3
	Part - B			
1.	Explain the X.509 certificate format. Describe its structure, fields, and the role it plays in ensuring secure communication in a public key infrastructure (PKI).	10	2	2
2.	Illustrate the working of Kerberos protocol with a complete example showing ticket generation, exchange, and verification steps. Also, mention how Kerberos ensures protection against replay attacks.	10	2	1
3.	What is a Certificate Authority (CA)? Explain how certificate issuance, validation, and revocation are handled in a PKI system. What are CRLs and OCSP?	10	3	3
4.	Describe the architecture and operation of IP Security (IPSec). Include its modes of operation, key protocols (AH and ESP), and the differences between them.	10	2	4
5.	Explain the IEEE 802.11 wireless LAN architecture. Discuss the components such as BSS, ESS, access point, and the role of the MAC and PHY layers.	10	2	3